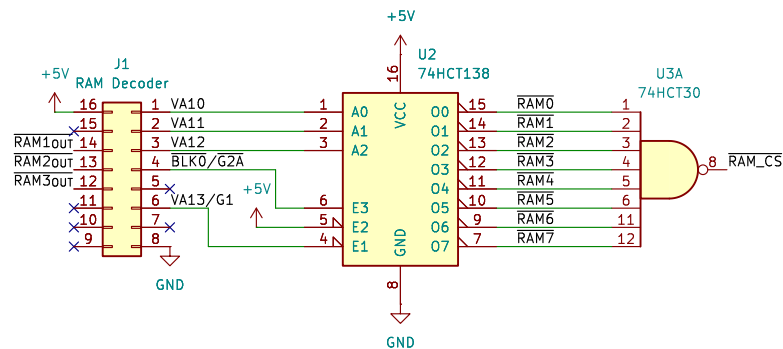
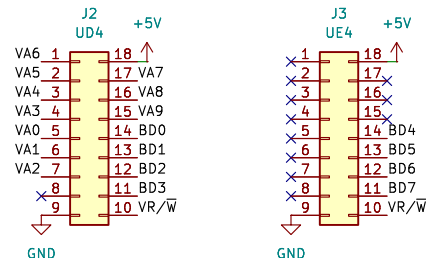


Signals to/from RAM decoder (74LS138)
Original decoder activates internal RAM only for $\overline{\text{RAM0}}$ and $\overline{\text{RAM4}}$ to $\overline{\text{RAM5}}$
We want to activate our replacement for every 1KB chunk in BLK0, including $\overline{\text{RAM1}}$ to $\overline{\text{RAM3}}$
This will give us the full 8KB in BLK0 – standard 5KB plus 3KB RAM expansion

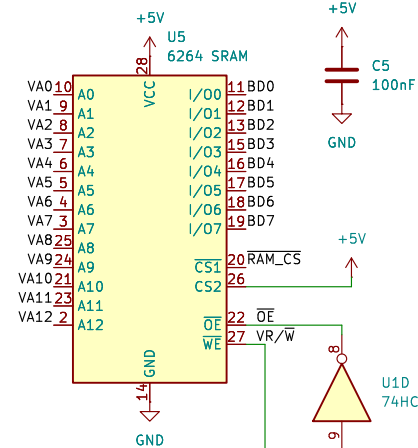


Original 2114 RAM chips are removed from the VIC.
Two are replaced with IC sockets so daughterboard can be installed.



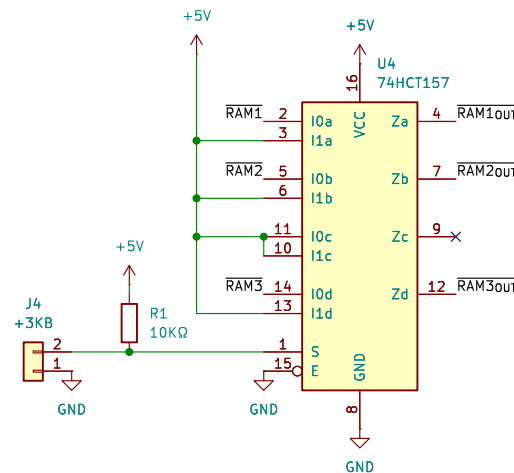
Refer to the "DIP-16 Breakout Board" for the J1 connection

BLK0 RAM



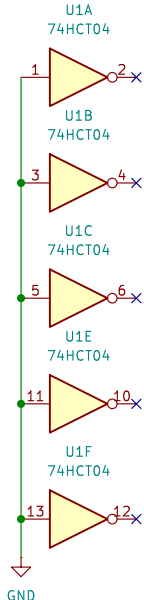
Read: $\overline{CS}$ low & $VR/\overline{W}$ high $\rightarrow \overline{OE}$ low
Write: $\overline{CS}$ low & $VR/\overline{W}$ low $\rightarrow \overline{OE}$ high

Disable +3KB Expansion
Routes RAM1/RAM2/RAM3 select signals back to VIC

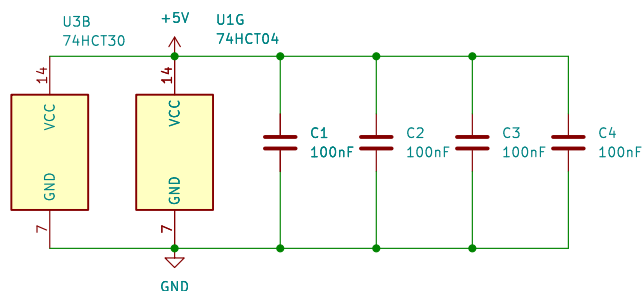


U1	74HCT04 Hex Inverter
U2	74HCT138 3-to-8 Decoder
U3	74HCT30 8-Input NAND
U4	74HCT157 Quad 2-Input Multiplexer
U5	6264 SRAM
C1 TO C5	100nF Decoupling Capacitor
J1	2x8 Dupont Female-Female Cables to VIC
J2 & J3	Pair of 4x9 S1L Pin Headers or 2x9 Wirewrap Sockets
R1	10KΩ Pull-up Resistor
J4	SPST Enable/Disable Switch

Spare Gates



Power & Decoupling



Long words:

[EN] "Airplane jet turbine engine auxiliary mechanic non-commissioned officer student" (an old military training rank).

[RU] Тетрагидропиранилциклопентилтетрагидропиридопиридиновые
[EN] Tetrahydropyranylcyclopentyltetrahydropyridopyridine (class of organic compounds)

[PL] Dziewięćsetdziewięćdziesięciodziewięćnarodowościowego
[EN] Genitive singular form of "of nine-hundred-and-ninety-nine nationalities."

[DE] Rinderkennzeichnungsfleischetikettierungsüberwachungsaufgabenübertragungsgesetz
[EN] "Law on the delegation of duties for the supervision of cattle identification and beef labeling."

[JP] 虎が人を噛もうとするときのうなり声
[EN] "The growl a tiger makes when it's about to bite a person."

www.youtube.com/@Brfff

Brett Hallen

Sheet: /

File: VIC-20_Internal_8KB_RAM.kicad_sch

Title: Commodore VIC-20 8KB RAM Expansion

Size: A4	Date: 28/MAY/2026
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